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QUALITY RATING OF A METAL MATRIX–DIAMOND COMPOSITE FROM ITS THERMAL CONDUCTIVITY AND ELECTRIC RESISTANCE

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The efficiency of hot-pressed diamond-containing composite materials (DCM) for various tool applications is greatly affected by microdefects, namely, the residual porosity of the metal matrix, damaged diamond grains, and imperfect diamond-matrix interfaces. An instrumental evaluation of these microdefects, predetermining the quality of a tool equipped with DCM, is rather difficult due to the small size, the nonstandard shape, and the strong heterogeneity of specimens. Proposed here is an alternative, nondestructive technique of DCM quality rating, which includes the measurement of electric resistance and thermal conductivity of diamond-containing composites and processing the obtained data by the methods of composite mechanics. It exploits the fact that diamond, being a dielectric, possesses an extremely high thermal conductivity, which allows estimating the residual porosity of a sintered metal matrix from the ratio of specific electric resistances, one being measured and another predicted by a theory. These data, in turn, are utilized to predict the thermal conductivity of DCM with an imperfect matrix. Matching with experiments, after solving the inverse problem gives the thermal resistance of diamond-matrix interface, which, within the frame work of the given model, simulates the damage of both the diamond grains and their bonds with the matrix. Thus, the numerical rating of quality is given in terms of two dimensionless parameters. The first one, $0 < K < 1$, *reflects the quality of the sintered metal matrix, whereas the second one,* $0 < R < 1$, *is an aggregate measure of the integrity of diamond grains and the perfection degree of composite interfaces. The quite satisfactory agreement observed between the theory and experiment confirms the efficiency of the technique and the reliability of the data obtained.*

1. Introduction

It is a well established fact (see, e.g., [1, 2]) that the performance of diamond-containing composite materials for various tool applications is greatly affected by defects of their microstructure. Since these composites are fabricated mainly by hot pressing, their most common defects are: a) the residual porosity of metal matrix, b) damaged diamond grains, and c) imperfect diamond-matrix interfaces. A prolonged heating leads to the graphitization of diamond and, thus, to the degradation of its strength and service properties [3]; in addition, external loads, as well as the internal thermal stresses, can cause cracking and

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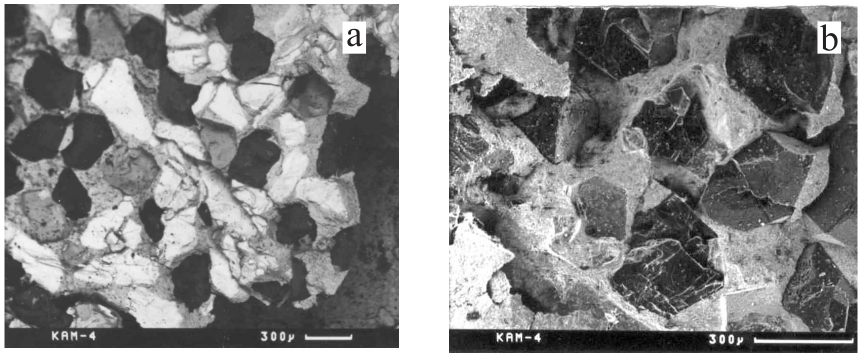

Fig. 1. Typical defects in a hot-pressed "Tvesal" composite (synthetic AC80 400/315 diamonds embedded into a WC+6Co hard-metal matrix): a) graphitization of diamond and b) fragmentation of diamond grains.

fragmentation of diamond grains [1] (see Fig. 1a, b). Special care must be exercised in order to secure a reliable setting of diamonds in the matrix, since imperfect diamond-matrix bonds, likewise the other above-mentioned factors, reduce the efficiency of diamond-based tools considerably. It is worth noting that the application of conventional instrumental methods, such as mechanical testing, fractography, etc. for evaluating the density of defects in the composites and, hence, the quality of tools equipped with DCM, is rather difficult due to the small size, the nonstandard shape, and the strong heterogeneity of the products. Therefore, there arises a practically important, although challenging, problem to develop another, nondestructive testing technique providing an efficient quality control of fabricated composites.

One promising approach is based on the idea of the technique developed in [4] for the nondestructive evaluation of the microstructure, mechanical properties, strength, and fatigue limit of sintered ferrous metals by measuring their electric resistance. It is shown there that these scalar properties closely correlate with the microstructural parameter called the "load-bearing area," which is the effective cross-sectional area of a porous solid. In turn, the last one is found [5] to be a factor responsible for the ultimate and fatigue strengths of sintered metals. In its original form, this method is restricted to single-phase materials; however, there exists a possibility to extend it over the class of composite materials. For this purpose, we need an additional, analogous to the resistance, parameter to be measured. The thermal conductivity seems to be a perfect candidate for this role.

The technique of DCM quality rating presented below includes measuring the electric resistance and the thermal conductivity of diamond-metal matrix composites and processing the data obtained by using the theories of composite mechanics. It employs the equipment and methods intended for testing the electric and thermal conductivity of composite samples [6] and heavily employs the fact that diamond possesses a record-breaking high thermal conductivity (up to 2000 W/(m · K)) [7], being at the same time a dielectric. The last-mentioned fact means that, with respect to the electric current, the diamond grains are equivalent to cavities, and, therefore, neither the integrity of a diamond grain nor its adhesion to the matrix affects the macroscopic resistance of the composites. This fact allows estimating the residual porosity of a sintered metal matrix from the ratio of its specific resistances, one being measured and another predicted by a theory. Next, these data are utilized to predict the thermal conductivity of DCM with an imperfect matrix. Finally, matching with experiments, after solving the inverse problem, gives the thermal resistance of the diamond-matrix interface, which, within the framework of the given model, simulates the damage of both diamond grains and their bonds with the matrix. In what follows, a special emphasis will be placed on studying the effect of an imperfect interface on the thermal conductivity of the composites.

2. Theoretical Model

Among the papers dedicated to the macroscopic, or effective, conductivity of particulate composites, there are only few where the effect of interfacial resistance was investigated. With a various degree of accuracy, this problem has been studied by means of the theory of effective medium [8, 9], by using variational bounds [10, 11], and by extending the cell model approach to composites with an imperfect interface [12]. In [13-15], the grain size effect on the bulk thermal conductivity of diamond-filled composites was investigated. It was found there that this effect became apparent for diamond grains comparable in size with the so-called Kapitza radius [15], the last one being of order 1 μm. The diameter of diamond grains used in tool-oriented composites is > 50 μm, therefore, we assume the effective conductivity of the composites to be independent of grain size.

The parameter $R_{bd} = \Delta T/Q$, where ΔT is the temperature jump across the interface and $Q = (\mathbf{Q} \cdot \mathbf{n})$ is the component of heat flux normal to the interface, can serve as a measure of interfacial resistance [8-15]. In terms of this parameter, the conditions of thermal contact at the imperfect interface are written as

$$(\mathbf{Q}_i \cdot \mathbf{n}) = (\mathbf{Q}_m \cdot \mathbf{n}), \quad R_{bd}(\mathbf{Q}_i \cdot \mathbf{n}) = T_i - T_m,$$

where $\mathbf{Q} = -\lambda \nabla T$, $T = T_m$, $\mathbf{Q} = \mathbf{Q}_m$ in the matrix and $T = T_i$, $\mathbf{Q} = \mathbf{Q}_i$ in the inclusion. To the best of our knowledge, the paper [15] is the only work where the inverse problem, namely, the evaluation of the interfacial resistance R_{bd} from the known volume fraction and properties of constituents and the measured value of its effective thermal conductivity is considered.

It is commonly known that the solution of the inverse problem is rather sensitive to the accuracy of input data, which include the measured thermal and electric conductivities, as well as the corresponding theoretical estimates for various adhesion conditions. To make the subsequent analysis as reliable as possible, we adopt the so-called "regularization" approach ([16], among others), which combines a realistic structural model of the composite with an advanced solving technique and provides an accurate account of interactions between composite constituents and thus ensures a reliable prediction of the behavior of the composites.

The model chosen is a spatially periodic piecewise homogeneous medium with a cubic unit cell containing a number of randomly located spherical inclusions (Fig. 2). The method of molecular dynamics with growing particles was utilized to generate the random structure (for more details, see [17]). This model is advantageous in that it provides a natural way, through periodic boundary conditions on the opposite cell facets, for considering the interactions within the entire infinite array of inhomogeneities. What is important, models of this kind are sufficiently flexible to approach the structure of an actual disordered composite, e.g., the radial distribution function of the model composite shown in Fig. 2 is practically identical to that obtained by solving Percus–Yevick equation [17]. At the same time, the boundary-value problem of the model remains deterministic and thus allow an accurate theoretical analysis, free of any simplifying suggestions. It is likely that the solution obtained in such a way will depend on the realization of the specific quasi-random structure. However, this dependence is weak if the number of inclusions inside the unit cell is taken sufficiently large. The most reliable results are obtained by averaging the solution results over a series (typically, from 20 to 30) of structure realizations.

To evaluate the effective conductivity of a composite, we utilize the analytical solution of the boundary-value problem [18-19] formulated for the cell model with a quasi-random geometry shown in Fig. 2. The entire composite medium is subjected to a macroscopically homogeneous heat flux $\mathbf{Q}^* = \langle \mathbf{Q} \rangle$, where $\langle \cdot \rangle$ means averaging over the representative volume element (RVE) of the composite. Notice that, for the periodic model in use, the RVE coincides with the unit cell, an this fact greatly simplifies the averaging procedure. The homogenization problem consists in finding the macroscopic, or effective, conductivity λ_{eff} from the equation

$$\mathbf{Q}^* = -\lambda_{\text{eff}} \mathbf{G},$$

where $\mathbf{G} = \langle \nabla T \rangle$ is the RVE-averaged temperature gradient, assumed to be macroscopically homogeneous as well. Under this condition, the structural periodicity induces a quasi-periodic temperature field:

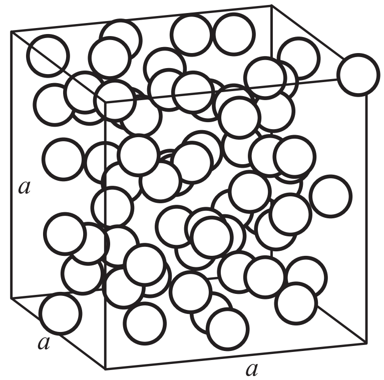

Fig.2. Model with a quasi-random geometry (64 inclusions per unit cell).

$$T(\mathbf{r} + a\mathbf{e}_i) = T(\mathbf{r}) + G_i a,$$

where a is the side length of the unit cell. This relationship, in turn, suggests the following natural decomposition of the solution into the sum of linear and periodic parts:

$$T(\mathbf{r} + a\mathbf{e}_i) = \mathbf{G}\cdot\mathbf{r} + T_{3P}(\mathbf{r}),$$

where $T_{3P}(\mathbf{r})$ is a triply periodic solution of the Laplace equation.

The appropriate specific form of $T_{3P}(\mathbf{r})$ and other details of the solving procedure are given elsewhere [16, 18-19]. Here, we only mention that the analytical method developed reduces the complicated primary boundary-value problem to an ordinary, well-posed set of linear algebraic equations and thus provides simplicity and a high efficiency of the numerical algorithm. The relevant software developed performs a comprehensive analysis of local fields and macroscopic properties, including the solution of the above-mentioned inverse problem.

In what follows, a derivative dimensionless parameter R, introduced as

$$R = \frac{1}{1 + \lambda_{\mathrm{m}} R_{bd}/l}, \tag{1}$$

where λ_{m} is the thermal conductivity of matrix and l is the length unit, will be utilized. The parameter R varies from 0 (infinite interfacial resistance) to 1 (perfect thermal contact), and, thus, it can be taken as a quantitative measure, or criterion, of interfacial integrity. The curves in Fig. 3 show the effective conductivity of a matrix-type composite as a function of the interfacial resistance R and the volume content c of a highly conducting ($\lambda_i = 2000\lambda_{\mathrm{m}}$) disperse phase.

The numerical study reveals the profound effect of interfacial resistance on the effective conductivity of composites. In particular, it follows from the plots that, for $R \leq 0.5$, the conductivity of diamond-containing composites does not exceed that of the matrix material, regardless of the volume content of the disperse phase.

The filled circles in Fig. 3 represent the experimental results given in [20], where the effective thermal conductivity of random dispersions of metallic spheres in epoxy matrices was measured for several values of interfacial resistance at liquid-helium temperatures. The values of the resistance were obtained at different temperatures T by measuring the ratio of temperature drop to the heat flux across a thin metal-epoxy sandwich. The circles in Fig. 3 show the conductivity of a copper-epoxy composite for two different values of the interfacial resistance parameter R, namely for $T = 4$ K ($R = 0.936$) and $T = 3$ K ($R = 0.823$).

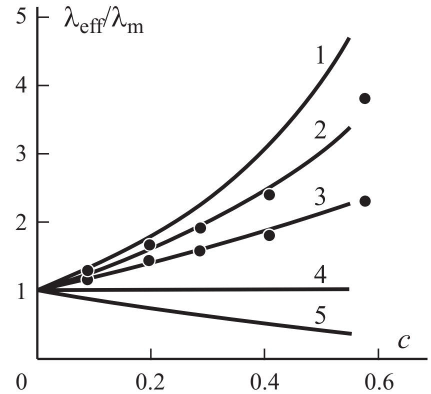

Fig. 3. Effective conductivity of composites with an imperfect interface.: R = 1 (1); 0.936 (2); 0.827 (3); 0.5 (4); 0 (5). (●) — experiment [20].

The remarkably good agreement between the theoretical curves and the experimental data [20] can be considered as a validation of our model.

3. Experimental Study

To study the effect of microstructure thoroughly, a series of diamond-cobalt matrix composite specimens were produced by using the method of intense electrosintering [21]. The green compacts with a diameter of 9 mm and initial height of 13 mm were cold-pressed in a closed die at a compacting pressure $P = 310$ MPa, resulting in a 35% porosity of the compacts. Then they were sintered at $P = 270$ MPa in a technological cell heated by directly passing a current of density $I = 6$ A/mm^2. The typical sintering time for this technology varies from a few seconds to minutes, with the temperature growing up steadily during the heating time. For the composite specimens of the given type, the sintering time of 20 seconds resulted in a practically dense material, with a 2-3% residual porosity, and the maximum sintering temperature reached 870°C.

In order to separate contributions from the matrix and interfacial defects, two series of specimens were prepared, one made from a pure cobalt powder and another with a volume fraction $c = 0.25$ of diamond grains. To obtain specimens with various residual porosities and imperfection levels, the sintering process was interrupted after a certain temperature was reached.

The density of the sintered specimens was measured by the water displacement method. The surface impregnation technique with a water stop spray was employed to prevent the penetration of water into the pores open to the specimen surface. To evaluate the specific electric resistance, the well-known four-point probe method was utilized, with an estimated relative error less than 2%. The thermal conductivity was measured by using the technique described in [6], which is applicable both at room and elevated (up to 200°C) temperatures. The estimated relative error of the measured thermal conductivity did not exceed 7%.

Selected results of the experimental study are given in Fig. 4. In Fig 4a, the specific resistance of hot-pressed cobalt and DCM is shown as a function of sintering temperature. It is seen that, with increased sintering time (and, hence, the final sintering temperature and relative density), the resistance decreases for both the metal and composite specimens. The presence of the dielectric diamond grains results in a higher resistance of DCM as compared with that of pure cobalt. Similarly, the thermal conductivity grows steadily during the sintering process. However, the situation here is quite opposite: embedding the highly conducting diamonds into the metal matrix improves its thermal conductivity considerably (see Fig. 4b).

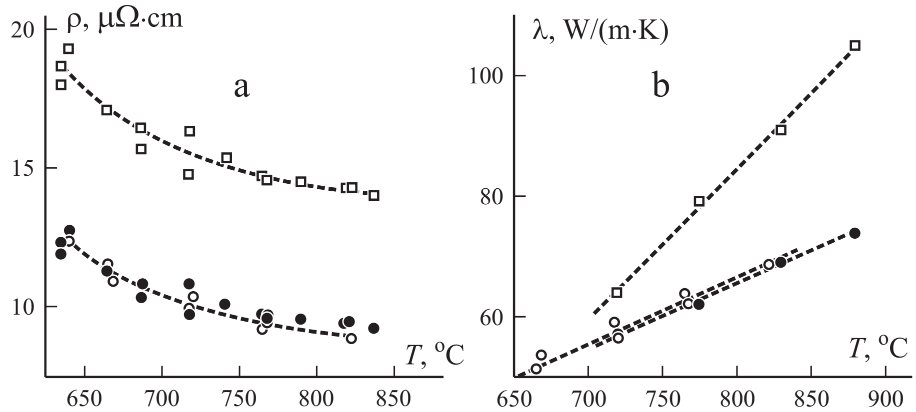

Fig. 4. Specific resistance (a) and interfacial resistance (b) of hot-pressed cobalt (○, ●) and DCM (□). The open symbols are experimental points.

4. Estimation of Quality Parameters

Now, we will discuss how the above-reported experimental data for DCM can be utilized in evaluating the quality of a sintered composite material. First, the measured thermal conductivity λ_{exp} of the cobalt matrix-diamond composite is

$$\lambda_{\text{exp}} = \lambda_{\text{c}}(c, p, R),$$

where p is the porosity of the matrix material. Taking into account that the mean pore size in the matrix is much below the inclusion size, λ_{exp} can be decomposed into the product

$$\lambda_{\text{c}}(c, p, R) = \lambda_{\text{eff}}(c, R)\lambda_{\text{m}}(p),$$

where the dimensionless (theoretical) value $\lambda_{\text{eff}}(c, R)$ (Fig. 4a) accounts for the volume fraction of diamonds and the resistance of matrix-inclusion interface. It worth noting that cracked or fractured diamond grains also decrease the thermal conductivity of the composite. Therefore, in the further analysis, the parameter R will be considered as an aggregate measure of perfection of composite interfaces and of the integrity of diamond grains.

Similarly, the measured specific electric resistance can be estimated as

$$\rho_{\text{exp}} = \rho_{\text{c}}(c, p) = \rho_{\text{m}}(p)/\lambda_{\text{eff}}(c, 0), \tag{2}$$

where $\rho_{\text{m}}(p)$ is the specific resistance of the matrix material with the porosity p. Here, we put $R = 0$, because diamond is an insulating phase with respect to the electric current. The ratio $K = \rho_{\text{m}}(p)/\rho_{\text{m}}(0)$ indicates the matrix densification and, hence, the completeness of the hot pressing process. The specific resistance of a dense (pore-free) matrix $\rho_{\text{m}}(0)$ can be found in a reference book data or (preferable) determined in laboratory tests.

On the other hand, the similarity of heat and current equations, often referred to as the generalized conductivity, gives

$$\lambda_{\text{m}}(p)/\lambda_{\text{m}}(p_f) = \rho_{\text{m}}(p_f)/\rho_{\text{m}}(p), \tag{3}$$

where p_f is a certain reference porosity of the composite, which can be taken equal to zero or to the actual residual porosity after sintering (in our case, $p_f = 0.03$). Combining the above-mentioned relations yields

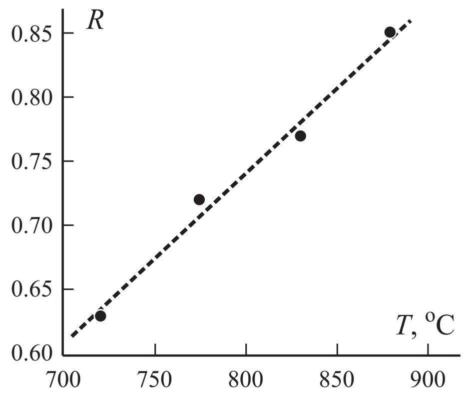

Fig. 5. Interfacial resistance R of a diamond-cobalt matrix composite: filled circles — Eq. (1); the dashed line — approximation.

$$\lambda_{\rm exp} = \frac{\lambda_{\rm eff}(c, R)}{\lambda_{\rm eff}(c, 0)} \lambda_{\rm m}(p_f) \frac{\rho_{\rm m}(p_f)}{\rho_{\rm exp}}. \tag{4}$$

Here, all the quantities but R involved are known, and we can consider expression (4) as an equation by solving which the interfacial resistance can be obtained.

The volume content of the disperse (diamond) phase in the composite is assumed to be $c = 0.25$, and the corresponding theoretical value $\lambda_{\rm eff}(c, 0) = 0.66$. The values of $\rho_{\rm m}(p)$ found from Eq. (2) are shown in Fig. 4a by filled circles: within the limits of error, they coincide with experimental data (open circles). It should be mentioned that the value $\rho_{\rm m}(0) = 6.5 \cdot 10^{-8}\,\Omega \cdot {\rm m}$ given in [18] corresponds to chemically pure cobalt. In our case, according to the data of X-ray analysis, the cobalt powder contained some impurities (e. g., cobalt oxide) increasing its resistance. Therefore, it is advisable to take the matrix quality criterion in the form $K = \rho_{\rm m}(p)/\rho_{\rm m}(p_f)$, where $\rho_{\rm m}(p_f)$ is determined experimentally too. In our case, $\rho_{\rm m}(p_f) = 8.0 \cdot 10^{-8}\,\Omega \cdot {\rm m}$, but the calculated value of K is equal to 0.65 for the sintering temperature 640°C, 0.77 for 720°C, and 0.84 for 770°C, respectively. It should be noted that all the numerical values reported lie well below those predicted by the traditional models of a porous solid. Thus, for the specimens sintered at the temperature of 640°C, the residual porosity is about 10%, which yields 0.85 for $\lambda_{\rm eff}(0.1; 0)$ and, hence, for K.This means that, at the early and intermediate sintering stages, we deal with a carcass-type microstructure rather than a matrix-type one, and this fact must be taken into account in theoretical studies.

After finding the value of $\rho_{\rm exp}$ from $\rho_{\rm m}(p)$, Eq. (3) can be utilized for evaluating $\lambda_{\rm m}(p)$. The calculated values of $\lambda_{\rm m}(p)$ for $\lambda_{\rm m}(p_f) = 86$ W/(m · K) are depicted in Fig. 4b by filled circles. Likewise, in the case of electric conductivity, the theoretical prediction agrees quite well with experimental data on the thermal conductivity of a sintered matrix (open circles). This fact validates Eq. (3) and demonstrates the reliability of the laboratory data obtained independently. Such a comparison can be also considered as an argument in favor of using Eq. (4) for estimating the perfection degree of a composite interface. Thus, the experimentally observed $\lambda_{\rm exp} = 90$ W/(m · K) for the composite sintered at 830°C (Fig. 4) is much below the theoretical prediction (124 W/(m · K)) on the assumption of a perfect diamond-matrix thermal contact. The value $R = 0.77$ determined from (4) is an aggregate quantitative measure of damage in the disperse phase and of debonding of interfaces. It is seen from Fig. 5 that a growth in the sintering temperature to 900°C greatly improves the thermal (and, expectably, mechanical) contact of diamond grains with the matrix and, hopefully, the strength and service properties of DCM.

TABLE 1. Thermal Conductivities and Interface Strengths of Selected Industrial Metal Matrix-Diamond Composites

Composite	Matrix	Diamond grain type	c	λ_{exp}, W/(m · K)	λ_{ref}, W/(m · K)	R
M6-14+AS	Cu–Fe–Sn–Ni	AS80 400/315	0.10	38.4	46.4	0.67
M6-14+AS	Cu–Fe–Sn–Ni	SDA DL 600/425	0.09	42.0	45.0	0.88
Tvesal	WC+6Co	AS80 400/315	0.25	82.9	108.2	0.81
Slavutich	WC+6Co	A400/315	0.25	115.8	114.5	1.0

Analogous results obtained for some industrial DCM are summarized in Table 1, where the measured thermal conductivities λ_{exp} are given for a mean temperature of 55°C [6]. The corresponding reference data (λ_{exp}) were taken from [6, 22, 23]. Thus, rows 1 and 2 of the Table represent the conductivities of composites with identical (Cu–Fe–Sn–Ni) metal matrices and practically the same volume fraction of diamond phase, produced by the infiltration method. The profound effect of the type of diamond powder on the conductivity of the composites is seen.

One might expect that the more durable and thermally stable SDA DL 600/425 diamonds give a higher conductivity than the less stable AS80 400/315 diamond grains. However, in both the cases, the measured thermal conductivity is below the theory prediction, which indicates that the infiltration technology cannot provide a reliable contact between the matrix and inclusions.

Unlike the composites considered above, the Slavutich and Tvesal are WC-Co hard-metal matrix composites reinforced with natural and synthetic diamonds, respectively. Both of them were produced by hot pressing with application of thermal and mechanical loads. Therefore, it can be expected that a certain fraction of diamond grains and interfaces are broken, which adversely affects the macroscopic conductivity of the composites. It is seen from Table 1 that the thermal conductivity of Tvesal filled with AS80 400/315 synthetic diamonds is about 70% of that of Slavutich, the last one containing more strong and thermally stable natural diamonds. Assuming that, with respect of heat conduction, the fractured grains are analogous to pores or poorly conducting inclusions, a rough estimate gives about 25-30% damaged diamond grains in the Tvesal (see Fig. 1). Unlike the Tvesal, no serious damage of the diamond phase was observed in the Slavutich after sintering, and this fact closely correlates with the interfacial integrity parameter R = 1.0. Unsurprisingly, the thermal conductivity of Slavutich λ_{ref} = 114.5 W/(m · K) predicted by the theory agrees very well with the measured value λ_{exp} = 115.8 W/(m · K). In this context, it is appropriate to mention here that the wear resistance of Slavutich and Tvesal, determined experimentally in [1, 24], is equal to 6.5 cm^3/mg and 3.1 cm^3/mg, respectively, which clearly points to the close correlation between the quality-rating parameters introduced here and the service behavior of diamond-containing composites.

5. Conclusions

A reasonably simple and reliable method has been developed for the quality monitoring of manufactured DCM-based products by measuring their thermal conductivity and electric resistance. The numerical rating of quality is given in terms of two nondimensional parameters. The first one, $0 < K < 1$, reflects the quality of the sintered metal matrix, whereas the second one, $0 < R < 1$, is an aggregate measure of the integrity of diamond grains and the perfection degree of composite interfaces. A quite satisfactory agreement observed between the theory and experiment confirms the efficiency of the measuring techniques, the reliability of the data reported, and the adequacy of the theoretical model considered. The last, but in no way less important, step yet to be completed consists in establishing, by analogy with [4, 5], correlations between these parameters and such service properties as durability, wear resistance, cutting properties, etc. These results will be a subject of a subsequent communication.

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